

Big Data Mining Final Report

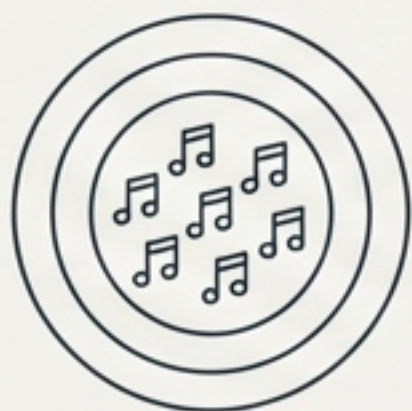
A Study on a Diverse Music Recommendation System
based on Stochastic MMR and Psychoacoustic Features

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The Challenge: Escaping the 'Filter Bubble' in Music Discovery

Traditional content-based recommenders, while accurate, often trap users in an "Echo Chamber," leading to monotonous recommendations and reduced long-term engagement.



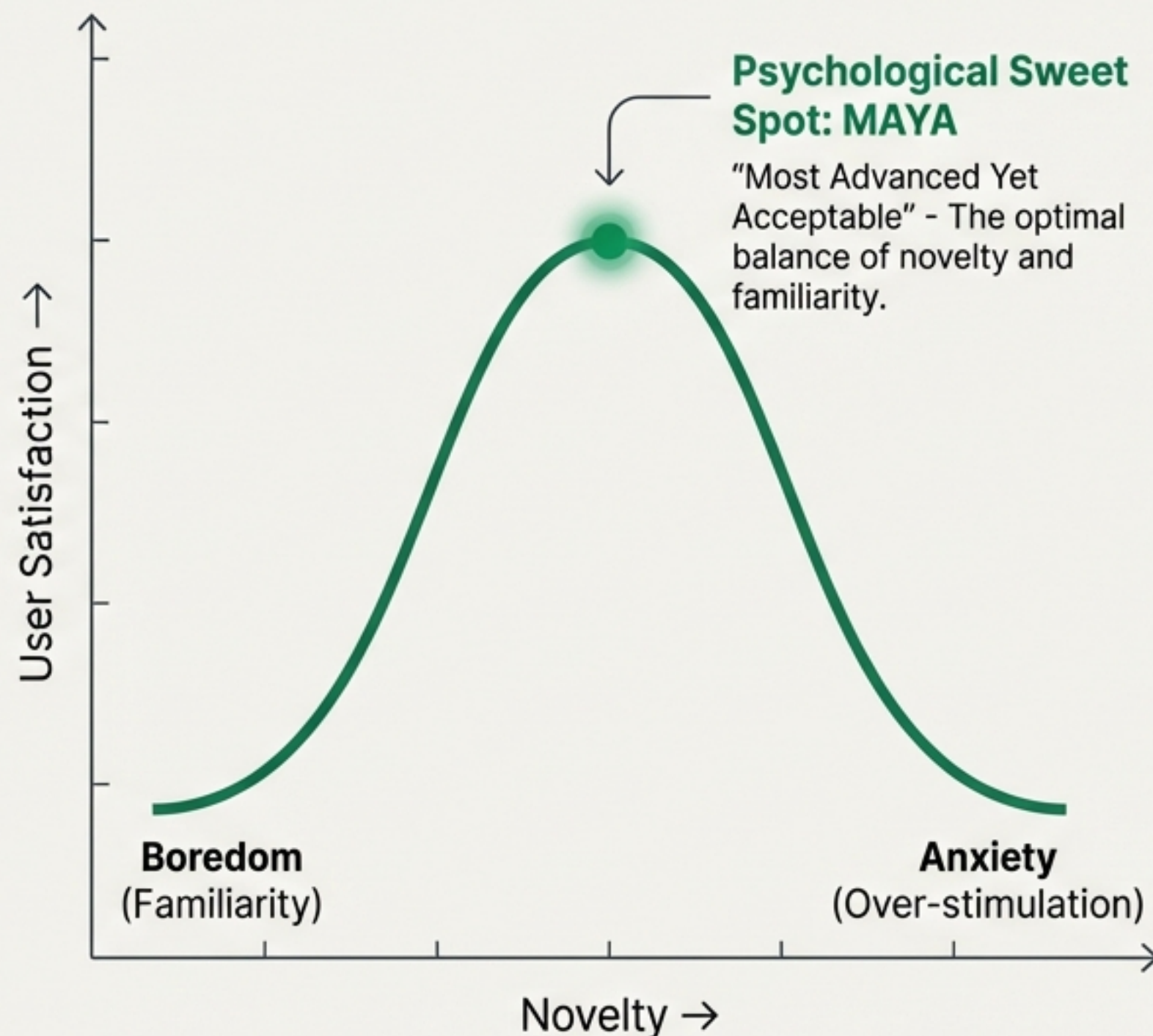
Filter Bubble

Over-specialization of recommendations that isolates users from diverse content.



Accuracy-Diversity Dilemma

The observation that optimizing solely for accuracy metrics often harms novelty. As noted by McNee et al. (2006), "Being accurate is not enough."



The Foundation: Deconstructing Music with Psychoacoustic Features

Dataset & Feature Engineering

Dataset: The Million Song Dataset (10,000-song subset)

Source: Columbia University LabROSA

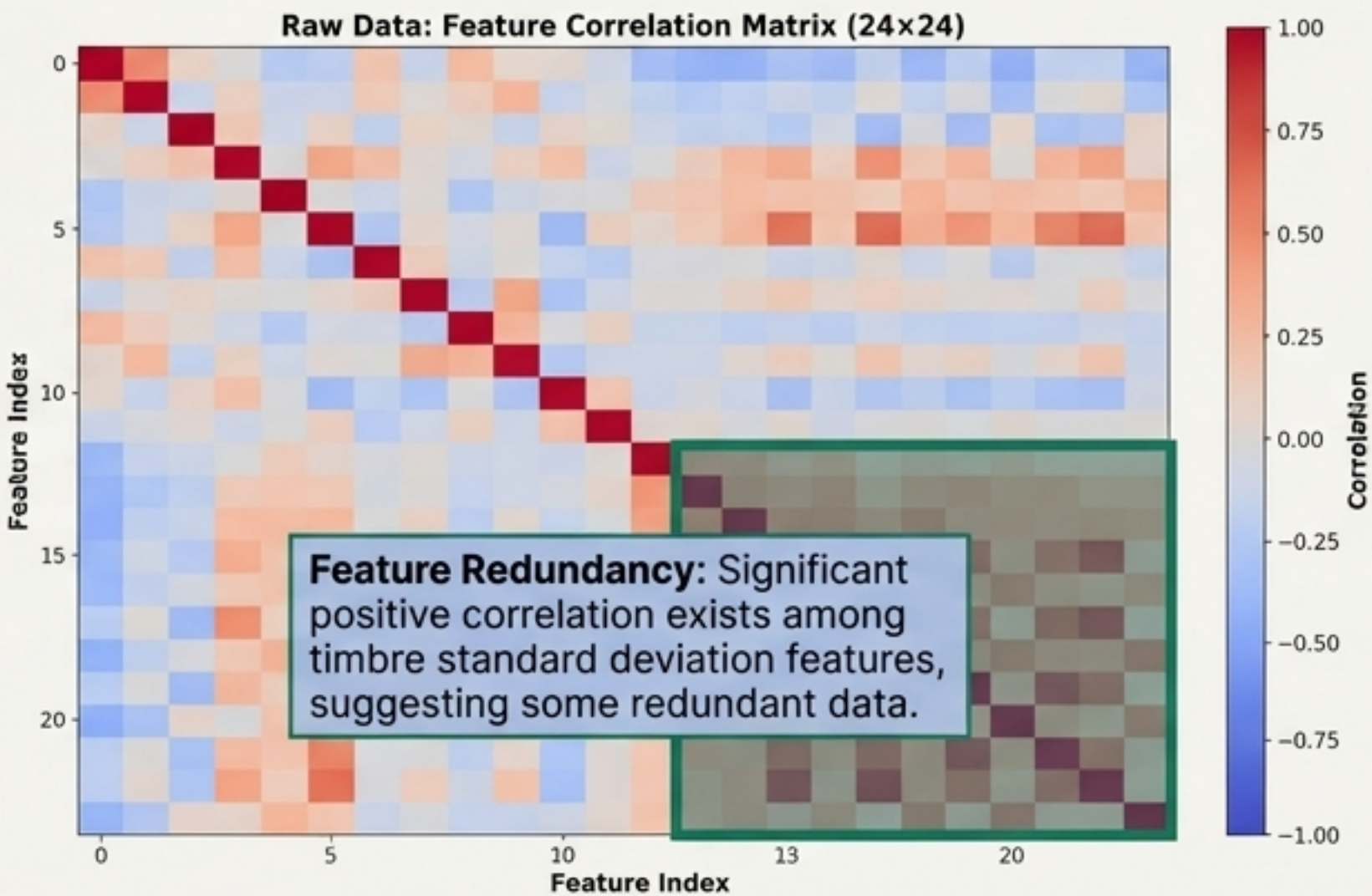
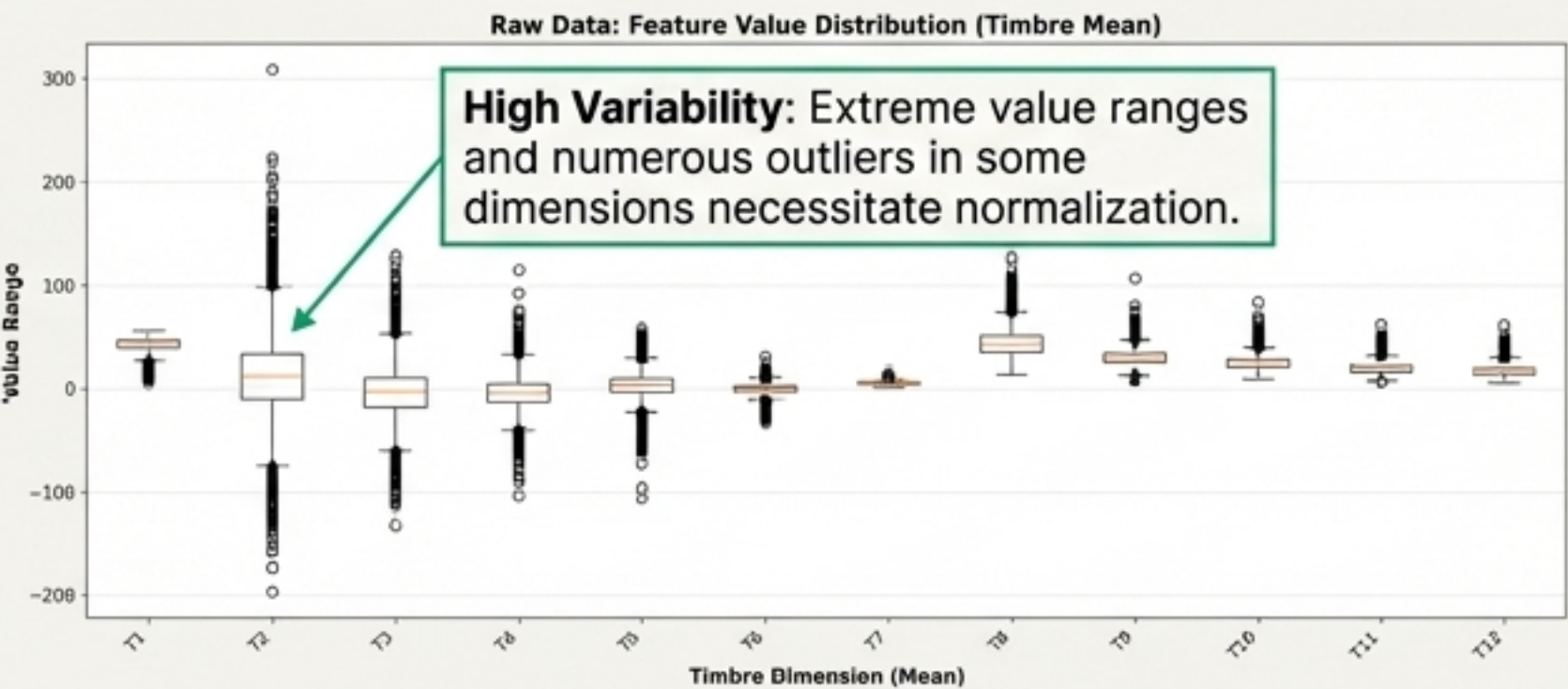
Data Quality: 100% feature extraction success rate; no NaN or Inf values.

Core Concept: Each song is represented as a **24-dimensional feature vector** derived from its **Timbre**, a key psychoacoustic feature.

$$feature_vector = [\mu_1, \dots, \mu_{12}, \sigma_1, \dots, \sigma_{12}]$$

(with μ_i defined as Mean and σ_i as Standard Deviation)

Exploratory Data Analysis Highlights



The Contenders: Five Approaches to Music Recommendation

A comparative study was designed to evaluate five distinct models, from standard baselines to diversity-focused algorithms.



Baseline Models (Accuracy-First)

1. **Pure Cosine Similarity:** A standard content-based method recommending the most mathematically similar items.
2. **Item-Item Collaborative Filtering (ItemCF):** Finds and recommends items similar to those a user has already liked.
3. **Matrix Factorization (MF-SVD):** Uses Truncated SVD to project a 10,000 x 10,000 similarity matrix into a 50-dimensional latent space.

Key Optimization: This method achieved an 87% reduction in memory usage.



Diversity-Enhanced Models

4. **Maximal Marginal Relevance (MMR):** The core proposed solution, designed to explicitly balance relevance with diversity.
5. **MMR + Temperature Sampling (MMR+Temp):** A stochastic variant of MMR that introduces randomness to enhance exploration and serendipity.

The Core Solution: How MMR Mathematically Engineers Diversity

MMR (Carbonell & Goldstein, 1998) reranks recommendations by iteratively selecting items that offer the best balance between relevance to the query and novelty compared to already selected items.

Relevance Term: Measures how similar a candidate song (s_i) is to the user's query song (q). This ensures accuracy.

$$s^* = \arg \max_{s_i \in C \setminus R} \left[\underbrace{\lambda \cdot \text{Sim}(s_i, q)}_{\text{Relevance Term}} - \underbrace{(1 - \lambda) \cdot \max_{s_j \in R} \text{Sim}(s_i, s_j)}_{\text{Redundancy Penalty Term}} \right]$$



Balancing Parameter λ : A tunable parameter that controls the trade-off. For this study, λ was set to 0.7, prioritizing high relevance while still enforcing diversity.

Redundancy Penalty Term: Measures the similarity of a candidate song (s_i) to its closest match in the *already recommended* list (R). This penalizes homogeneity.

Defining Success: A Unified Framework for Evaluation

To conduct a comprehensive and fair comparison, a unified evaluation framework with four distinct metrics was established.

The Four Pillars of Evaluation



Relevance

The average similarity between the recommended songs and the original query song. *Measures accuracy.*



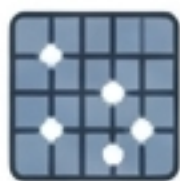
Diversity (ILD)

The average dissimilarity between all pairs of songs in the recommendation list. *Measures variety.*



Novelty

The ability of the system to recommend less popular, "long-tail" items. *Measures serendipity.*



Coverage

The percentage of the total song catalog that the system is able to recommend. *Measures utility of the database.*

The Weighted Score: Prioritizing Our Goal

To align the final ranking with the project's primary goal of overcoming the filter bubble, a weighted scoring system was created.

$$\begin{aligned} \text{Score} = & \textbf{(0.35 * Diversity)} \\ & + \textbf{(0.25 * Relevance)} \\ & + \textbf{(0.25 * Novelty)} \\ & + \textbf{(0.15 * Coverage)} \end{aligned}$$

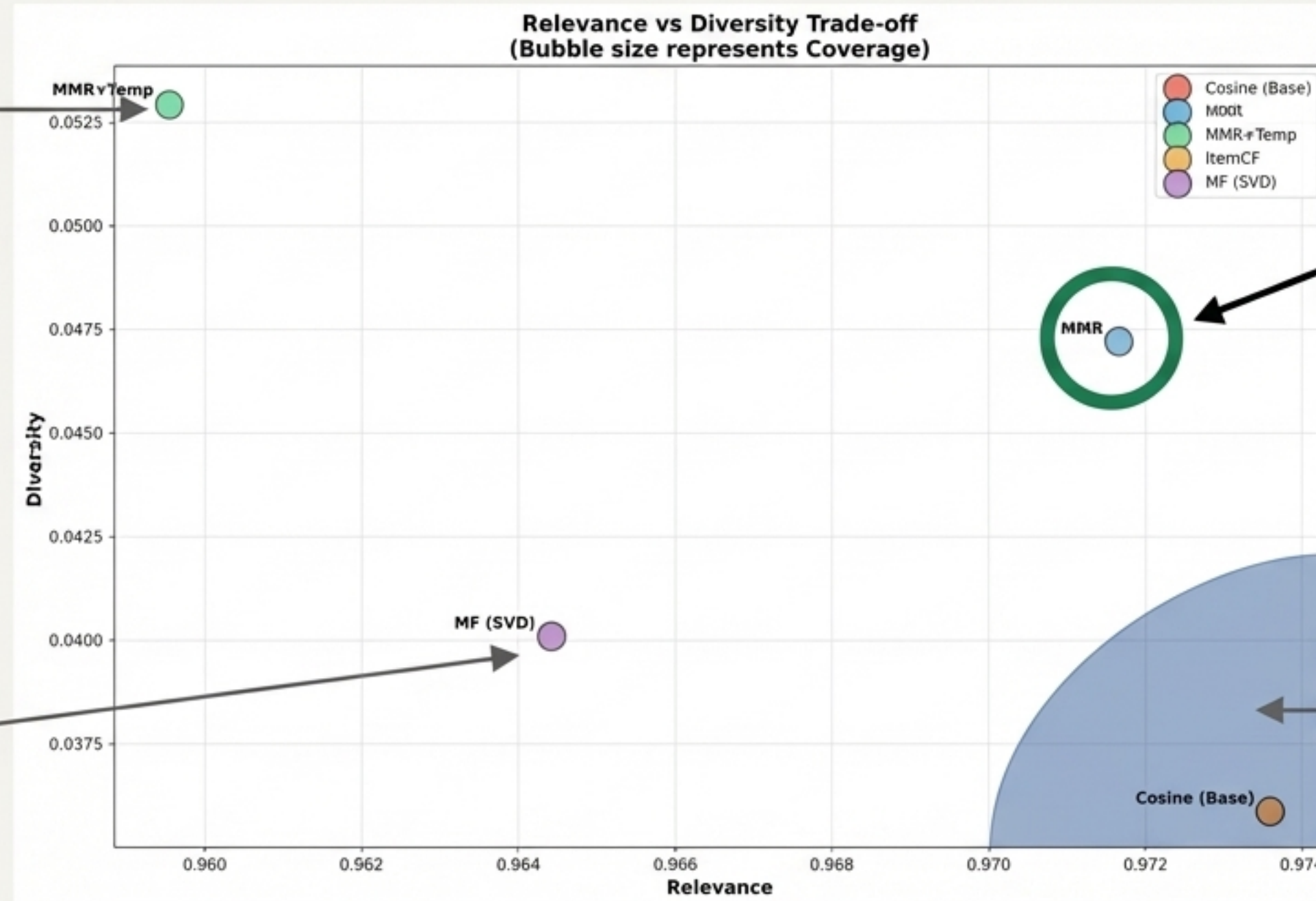
The decision was made to assign **Diversity the highest weight (35%)**, reflecting its central importance to the project's success.

The Central Trade-off: Charting Relevance vs. Diversity

This plot visualizes the performance of each algorithm, revealing the inherent tension between recommendation accuracy (Relevance) and variety (Diversity).

Top Left (High Diversity, Lower Relevance):
Stochastic sampling maximizes diversity at the cost of some relevance.

Also improves diversity, but is less relevant than MMR.

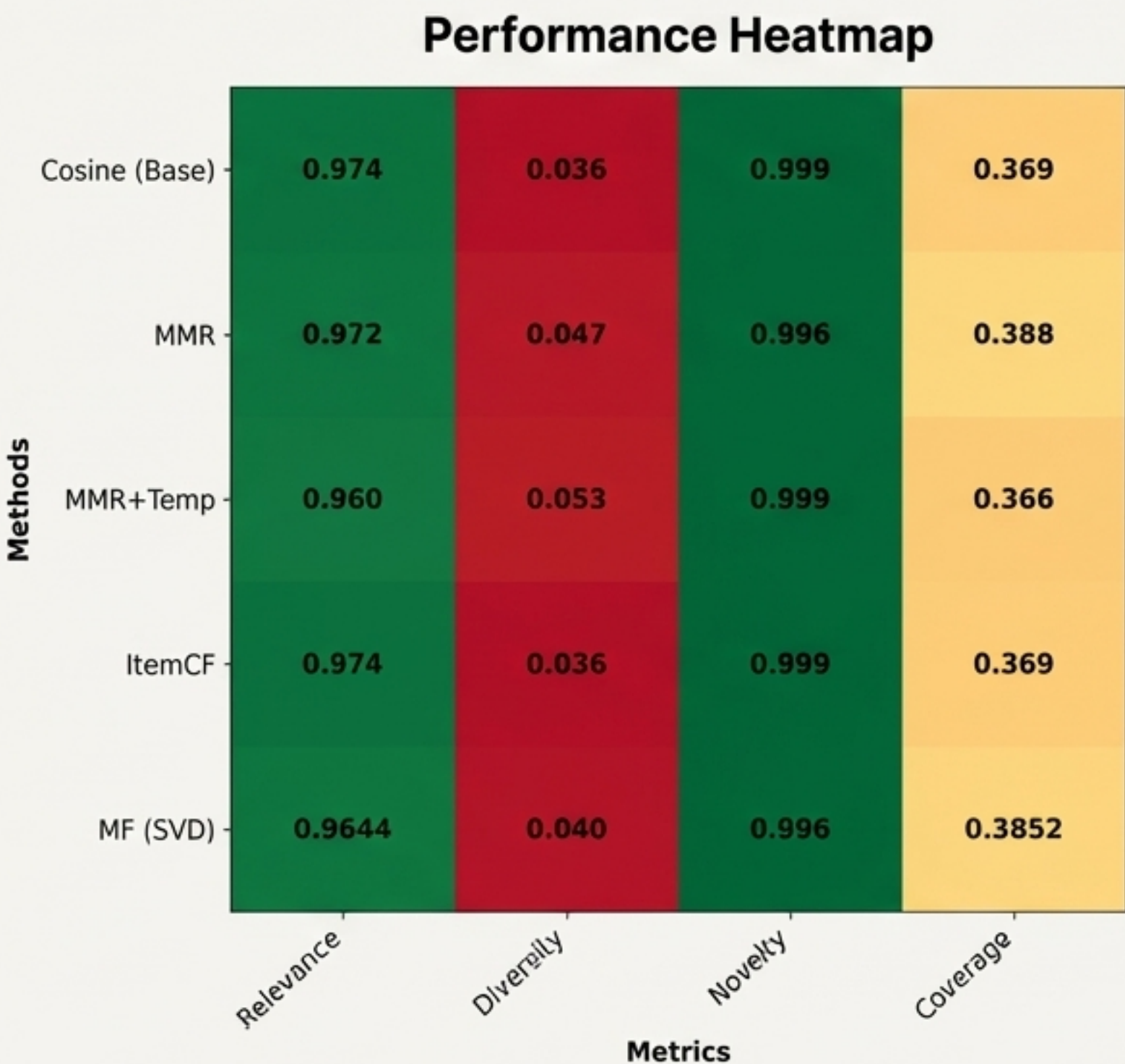


The Sweet Spot:
MMR achieves a significant boost in diversity while maintaining relevance nearly as high as the baselines.

Bottom Right (High Relevance, Low Diversity):
Baseline models are highly accurate but produce homogenous lists.

Results: Quantifying the Performance of Each Method

A detailed comparison across all four metrics shows a clear performance hierarchy, with diversity as the key differentiator.



Key Quantitative Finding

+31.73%

Increase in Diversity

-0.20%

Sacrifice in Relevance

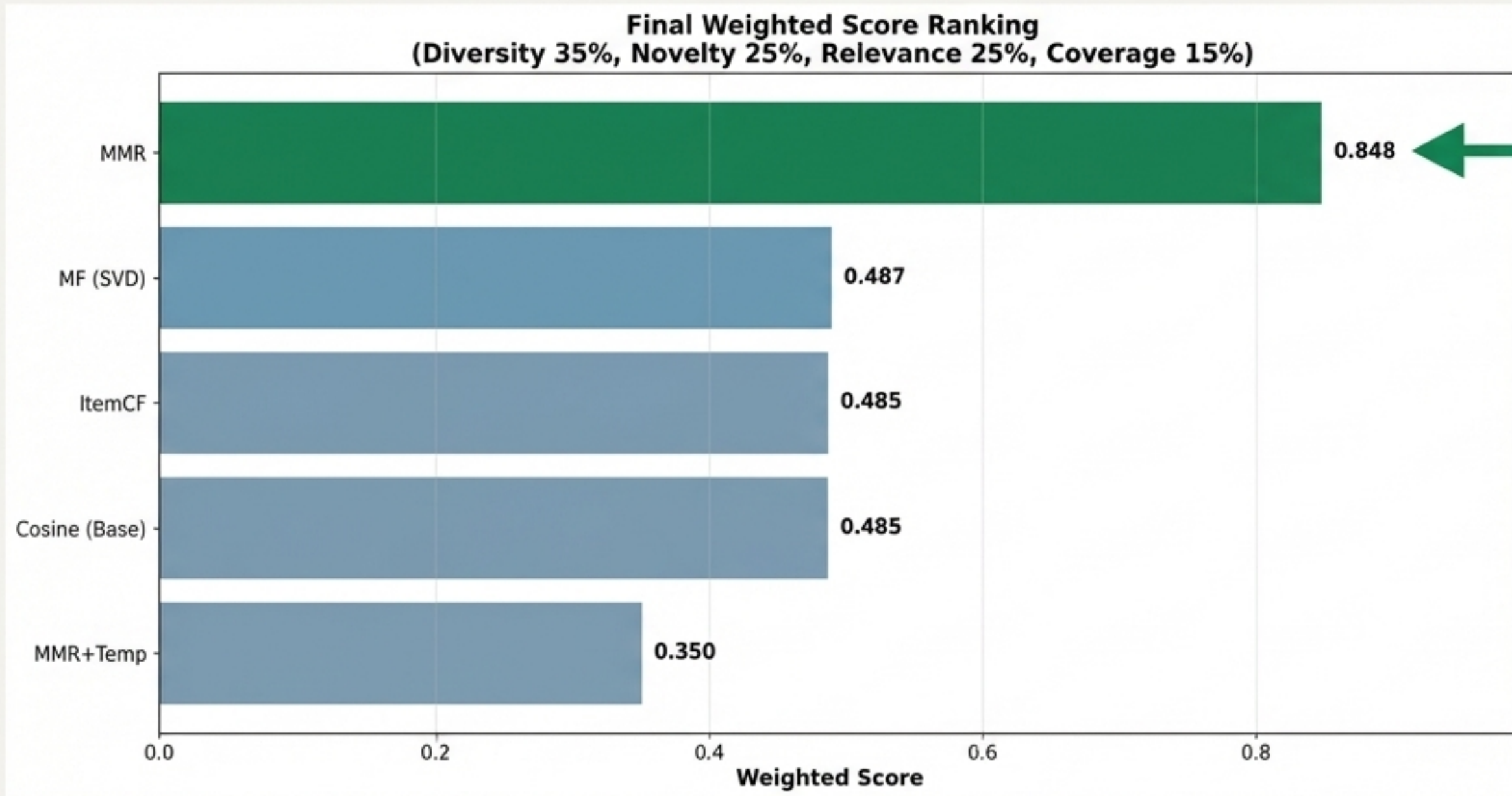
MMR vs. Cosine (Base) model

Full Results Table

Method	Relevance	Diversity	Novelty	Coverage
Cosine (Base)	0.9736	0.0359	0.9986	0.3691
MMR	0.9717	0.0472	0.9986	0.3882
MMR+Temp	0.9595	0.0529	0.9985	0.3656
ItemCF	0.9736	0.0359	0.9986	0.3691
MF (SVD)	0.9644	0.0401	0.9986	0.3852

The Verdict: MMR is the Decisive Winner

When evaluated against the unified weighted scoring framework—which prioritizes diversity—the Maximal Marginal Relevance (MMR) model emerges as the unequivocal top performer.



Final Score: 0.848

Substantially higher than the next best competitor, MF (SVD), at 0.487.

For the goal of building a recommendation system that balances accuracy with a significant improvement in diversity, MMR is the superior solution.

Conclusion and Future Directions

Key Achievements

- ✓ **1. Successfully Implemented and Compared 5 Models:** Ranging from industry-standard baselines to advanced diversity-enhancing algorithms.
- ✓ **2. Established a Comprehensive Evaluation Framework:** Moving beyond simple accuracy to measure diversity, novelty, and coverage.
- ✓ **3. Demonstrated MMR's Efficacy:** Proved that MMR can significantly increase recommendation diversity (**+31.7% green**) with a negligible impact on relevance (**-0.2%**).

Limitations & Next Steps

The study was conducted on a 10,000-song subset, used only timbre features, and relied on offline evaluation without user interaction data.

Future Work (Short-Term):

- Integrate user listening data
- Implement deep learning models (e.g., Neural CF)
- Add more feature types (lyrics, genre tags)

Future Work (Long-Term):

- Develop a hybrid recommender
- Implement A/B testing for live user feedback
- Explore context-aware recommendations (e.g., time, mood)